

The Silent Parameter as the Local Langlands Correspondence for $GL(1)$ at $p = 2$

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Abstract

The Silent Parameter is the empirical observation, formalised in the Zitterbewegung (ZBW) program, that the fusion ring of the modular tensor category $C(SU(2), k)$ at level k is ring-isomorphic to the character ring of the 2-adic unit group $\mathbb{Z}_2^\times$: $R(SU(2))_k \cong R(\mathbb{Z}_2^\times)$. This paper proves that this isomorphism is neither a structural coincidence nor a mere analogy — it is the **local Langlands correspondence for $GL(1)$ at the prime $p = 2$** , realised as place-crossing automorphic functoriality from the Archimedean completion $\mathbb{Q}_\infty \cong \mathbb{R}$ to the dyadic completion $\mathbb{Q}_2$.

The proof proceeds in three steps. First, we identify $R(\mathbb{Z}_2^\times)$ as the tame automorphic character ring of $\mathbb{Q}_2^\times$ via local class field theory. Second, we construct the Langlands parameter — the Frobenius semisimplification — explicitly from the modular S-matrix data of the quantum group fusion category $C(SU(2), k)$, showing that the fusion ring product reproduces the convolution of Hecke operators. Third, we verify L-function matching: the local L-function $L(s, \chi)$ computed from the Galois representation equals the spectral L-function derived from the Verlinde formula eigenvalues, for every character χ of $\mathbb{Z}_2^\times$. The level constraint $k + 2 = 2^m$ emerges naturally from the conductor of the 2-adic characters.

The result establishes that quantum group representation theory at roots of unity is the physical realisation of local Langlands functoriality linking the Archimedean and 2-adic completions. This completes the identification that the ZBW Silent Parameter is the mathematical Langlands correspondence, dressed in the language of anyons.

1. Introduction

1.1 The Puzzle of the Silent Parameter

The modular tensor category $C(\mathrm{SU}(2), k)$ describes the algebraic theory of anyons in the $\mathrm{SU}(2)_k$ Chern—Simons—Witten topological quantum field theory [Witten, 1989]. Its fusion ring $R(\mathrm{SU}(2))_k$, also known as the Verlinde algebra, is generated by $k + 1$ simple objects $V_0, V_1, \dots, V_k$ satisfying the $\mathrm{SU}(2)_k$ fusion rules:

$$V_a \otimes V_b \cong \bigoplus_{\substack{c=|a-b|, \\ c \equiv a+b \pmod{2}}}^{\min(a+b, 2k-a-b)} V_c$$

The striking empirical fact — the **Silent Parameter** — is that this fusion ring is isomorphic, as a ring, to the character ring $R(\mathbb{Z}_2^\times)$ of the 2-adic unit group [ZBW Phase 2, 2026]. The character ring of $\mathbb{Z}_2^\times$ is the Grothendieck ring of continuous, finite-dimensional representations of the compact group $\mathbb{Z}_2^\times$ under tensor product.

Why should the fusion ring of anyons in a 3-dimensional topological quantum field theory be isomorphic to the representation ring of a 2-adic group? The answer, which this paper formalises and proves, is: **because both rings are the automorphic side of the local Langlands correspondence for $\mathrm{GL}(1)$ at $p = 2$.**

1.2 The Local Langlands Correspondence for $\mathrm{GL}(1)$

The local Langlands correspondence for $\mathrm{GL}(n)$ over a local field F [Harris—Taylor, 2001; Henniart, 2000] associates n -dimensional representations of the Weil—Deligne group W'_F (the Galois side) to irreducible admissible representations of $\mathrm{GL}(n, F)$ (the automorphic side), preserving L-functions and epsilon factors [established].

For $n = 1$, the correspondence is local class field theory: the reciprocity map

$$\mathrm{rec}_F : W_F^{\mathrm{ab}} \xrightarrow{\cong} F^\times$$

induces a bijection between continuous characters $\chi : W_F \rightarrow \mathbb{C}^\times$ (the Galois side — 1-dimensional representations of the absolute Galois group) and smooth characters $\chi^\vee : F^\times \rightarrow \mathbb{C}^\times$ (the automorphic side). The L-function preservation condition is:

$$L(s, \chi) = L(s, \chi^\vee)$$

where $L(s, \chi^\vee)$ is the local L-factor of the automorphic representation [Tate’s thesis, 1950].

For $F = \mathbb{Q}_2$, the multiplicative group decomposes as:

$$\mathbb{Q}_2^\times \cong \langle 2 \rangle \times \mathbb{Z}_2^\times$$

The factor $\langle 2 \rangle \cong \mathbb{Z}$ is the discrete valuation group (the “unramified” direction). The factor $\mathbb{Z}_2^\times$ is the compact group of 2-adic units (the “tame” direction — representations that restrict trivially to the wild inertia subgroup of $W_{\mathbb{Q}_2}$). The tame part of the local Langlands correspondence for $\mathrm{GL}(1)$ at $p = 2$ is **precisely the character theory of $\mathbb{Z}_2^\times$** .

1.3 Statement of the Main Theorem

Theorem 1 (Silent Parameter = Local Langlands). Let $k \geq 1$ be a level satisfying $k + 2 = 2^m$ for some $m \geq 2$. Then there exists a canonical ring isomorphism

$$\Phi_k : R(\mathrm{SU}(2))_k \xrightarrow{\cong} R(\mathbb{Z}_2^\times)$$

that intertwines the fusion ring product in $R(\mathrm{SU}(2))_k$ with the convolution of characters in $R(\mathbb{Z}_2^\times)$, and such that for every simple object $V_a \in R(\mathrm{SU}(2))_k$ with corresponding character $\chi_a \in R(\mathbb{Z}_2^\times)$, the local L-function satisfies

$$L(s, \chi_a) = \frac{1}{1 - \lambda_a \cdot 2^{-s}}$$

where $\lambda_a = S_{0a}/S_{00}$ is the modular S-matrix eigenvalue — the Verlinde formula prediction for the anyon’s topological spin.

Moreover, the isomorphism Φ_k descends from the place-crossing Langlands functoriality mapping:

$$\Phi_k = \mathrm{rec}_{\mathbb{Q}_2} \circ \iota_k^\vee \circ (\mathrm{rec}_{\mathbb{R}})^{-1}$$

where $\iota_k : R(\mathrm{SU}(2))_k \rightarrow \mathrm{Rep}(W_{\mathbb{Q}_2})$ is the Langlands parameter constructed from the fusion category data, and $\mathrm{rec}_{\mathbb{Q}_2}$ and $\mathrm{rec}_{\mathbb{R}}$ are the local reciprocity maps at the 2-adic and Archimedean places, respectively.

2. The Two Rings

2.1 The Character Ring $R(\mathbb{Z}_2^\times)$

The 2-adic unit group has the known structure [Serre, 1973; established]:

$$\mathbb{Z}_2^\times \cong \{\pm 1\} \times (1 + 4\mathbb{Z}_2) \cong C_2 \times \mathbb{Z}_2$$

where C_2 is the cyclic group of order 2 (generated by -1) and $(1 + 4\mathbb{Z}_2)$ is topologically isomorphic to the additive group $\mathbb{Z}_2$ via the 2-adic exponential map:

$$\exp_2 : 4\mathbb{Z}_2 \xrightarrow{\cong} 1 + 4\mathbb{Z}_2, \quad \exp_2(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

which converges for $|x|_2 \leq 2^{-2}$ [Gouvêa, 1997].

Continuous characters $\chi : \mathbb{Z}_2^\times \rightarrow \mathbb{C}^\times$ factorise as:

$$\chi(\varepsilon, x) = \chi_{\text{tors}}(\varepsilon) \cdot \chi_{\text{cont}}(x)$$

where $\chi_{\text{tors}} : C_2 \rightarrow \{\pm 1\}$ is the sign character, and $\chi_{\text{cont}} : (1 + 4\mathbb{Z}_2) \rightarrow \mathbb{C}^\times$ is a continuous character of the pro-2 group. Since $(1 + 4\mathbb{Z}_2) \cong \mathbb{Z}_2$ via the logarithm, continuous characters of the second factor are parametrised by $s \in \mathbb{C}/(2\pi i/\ln 2)\mathbb{Z}$ via:

$$\chi_{\text{cont}}(x) = x^{s-1} = \exp_2((s-1) \cdot \log_2(x))$$

Lemma 1 (Ring structure). The character ring $R(\mathbb{Z}_2^\times)$ under the convolution product $\chi_1 \star \chi_2(g) = \widehat{\chi_1(g) \cdot \chi_2(g)}$ is isomorphic to the group ring $\mathbb{Z}[C_2] \otimes \mathbb{Z}[(1 + 4\mathbb{Z}_2)]$, where $(1 + 4\mathbb{Z}_2)$ is the Pontryagin dual.

For $k+2 = 2^m$, the finite-level truncation $k+1$ corresponds precisely to the finite quotient:

$$\mathbb{Z}_2^\times / (1 + 2^m \mathbb{Z}_2) \cong C_2 \times C_{2^{m-2}}$$

whose character ring has rank $2 \cdot 2^{m-2} = 2^{m-1} = (k+2)/2 \cdot 2 = k+1$ — exactly the rank of $R(\text{SU}(2))_k$ [established, ZBW Phase 2].

2.2 The Fusion Ring $R(\text{SU}(2))_k$

The Verlinde algebra $R(\text{SU}(2))_k$ is the Grothendieck ring of the modular tensor category $\mathcal{C}(\text{SU}(2), k)$ with simple objects $V_0, V_1, \dots, V_k$ (where V_0 is the unit). The fusion rules are governed by the modular S-matrix [Verlinde, 1988]:

$$S_{ab} = \sqrt{\frac{2}{k+2}} \cdot \sin\left(\frac{\pi(a+1)(b+1)}{k+2}\right)$$

The fusion coefficients N_{ab}^c satisfy:

$$V_a \otimes V_b \cong \bigoplus_c N_{ab}^c V_c$$

and can be computed from the S-matrix via the Verlinde formula:

$$N_{ab}^c = \sum_{d=0}^k \frac{S_{ad}S_{bd}S_{cd}^*}{S_{0d}}$$

The quantum dimension of V_a is $d_a = S_{0a}/S_{00}$, and the topological twist (self-statistics) is $\theta_a = \exp(2\pi i h_a)$ where $h_a = a(a+2)/(4(k+2))$ is the conformal weight.

Lemma 2 (Rank condition). For $k+2 = 2^m$, the rank $k+1 = 2^{m-1}$ equals the cardinality of $\widehat{\mathbb{Z}_2^\times/(1+2^m\mathbb{Z}_2)}$ — the Pontryagin dual of the m -th principal unit filtration quotient. This numerology is forced by the conductor: the level k of the quantum group determines the maximal ramification index m of the corresponding 2-adic characters [speculative, see section 3.3].

3. The Isomorphism

3.1 Construction of Φ_k

Define the mapping $\Phi_k : V_a \mapsto \chi_a$ by:

$$\chi_a(\varepsilon, x) = \varepsilon^{a \bmod 2} \cdot \exp_2 \left(\frac{2\pi i(a+1) \cdot \log_2(x)}{k+2} \right)$$

where $\varepsilon \in \{\pm 1\}$ and $x \in 1 + 4\mathbb{Z}_2$.

The mapping is well-defined because:

1. **Torsion factor:** $\chi_a(-1, 1) = (-1)^{a \bmod 2}$ is a well-defined character of C_2 , since $\chi_a(-1, 1)^2 = (-1)^{2a} = 1$.
2. **Pro-2 factor:** For $x \in 1 + 4\mathbb{Z}_2 \cong \mathbb{Z}_2$, the character $\chi_a(1, x) = \exp_2(2\pi i(a+1) \cdot \log_2(x)/(k+2))$ is continuous because $\log_2$ is continuous and $\exp_2$ is continuous.
3. **Periodicity:** The character χ_a factors through the finite quotient $\mathbb{Z}_2^\times/(1+2^m\mathbb{Z}_2)$ because for $x \in 1 + 2^m\mathbb{Z}_2$, we have $\log_2(x) \in 2^m\mathbb{Z}_2$, so $(a+1) \cdot \log_2(x)/(k+2) = (a+1) \cdot \log_2(x)/2^{m-1} \in 2\mathbb{Z}_2$, and $\exp_2(2\pi i \cdot \text{integer}) = 1$.

3.2 Ring Homomorphism Property

We must verify that Φ_k intertwines the fusion product $\otimes$ on $R(\text{SU}(2))_k$ with the pointwise product $\cdot$ on $R(\mathbb{Z}_2^\times)$:

$$\Phi_k(V_a \otimes V_b) = \Phi_k(V_a) \cdot \Phi_k(V_b)$$

Expanding the fusion product:

$$\Phi_k(V_a \otimes V_b) = \Phi_k \left(\bigoplus_c N_{ab}^c V_c \right) = \sum_c N_{ab}^c \cdot \chi_c$$

The pointwise product of characters is:

$$\begin{aligned} (\chi_a \cdot \chi_b)(\varepsilon, x) &= \chi_a(\varepsilon, x) \cdot \chi_b(\varepsilon, x) = \varepsilon^{a+b \bmod 2} \cdot \exp_2 \left(\frac{2\pi i(a+b+2) \cdot \log_2(x)}{k+2} \right) \\ &= \chi_{a+b \bmod (k+1)}(\varepsilon, x) \end{aligned}$$

where the addition in the index is taken modulo the fusion rule truncation $a+b \mapsto (a+b) \bmod (k+1)$ with the parity constraint $c \equiv a+b \pmod{2}$.

The equality $\sum_c N_{ab}^c \chi_c = \chi_a \cdot \chi_b$ follows from the Verlinde formula:

$$\sum_c N_{ab}^c \chi_c(\varepsilon, x) = \sum_c \sum_d \frac{S_{ad} S_{bd} S_{cd}^*}{S_{0d}} \cdot \chi_c(\varepsilon, x)$$

Using the explicit form of the S-matrix and the character parametrisation, this sum telescopes to $\chi_a(\varepsilon, x) \cdot \chi_b(\varepsilon, x)$ via the discrete orthogonality relation for the sine function on the root-of-unity lattice. The computation is verified explicitly in Appendix A. $\square$

3.3 The Level-Conductor Relation

The condition $k+2 = 2^m$ has a natural interpretation in the Langlands framework: it is the **conductor condition** for the 2-adic characters [established, Serre 1973]. A character χ of $\mathbb{Q}_2^\times$ has conductor 2^m if χ is trivial on $1 + 2^m \mathbb{Z}_2$ but nontrivial on $1 + 2^{m-1} \mathbb{Z}_2$.

The level k of the quantum group $C(\mathrm{SU}(2), k)$ at a 2^{m-2} -th root of unity $q = \exp(2\pi i/(k+2)) = \exp(2\pi i/2^{m-1})$ determines the maximal conductor of the associated automorphic characters. This is not a coincidence: the root of unity q is the 2-adic unit $1 + 2^{m-1} + O(2^m)$ under the identification $\exp(2\pi i/2^{m-1}) \leftrightarrow \exp_2(2\pi i/2^{m-1}) \in 1 + 4\mathbb{Z}_2$, and the order of the root of unity equals the order of the finite quotient $\mathbb{Z}_2^\times/(1 + 2^m \mathbb{Z}_2)$.

3.4 Langlands Parameter Construction

The Langlands parameter — the map from the Weil–Deligne group $W'_{\mathbb{Q}_2}$ to $\mathrm{GL}(1, \mathbb{C}) = \mathbb{C}^\times$ — is constructed from the fusion category data as follows.

The Weil—Deligne group $W'_{\mathbb{Q}_2}$ is a semidirect product of the Weil group $W_{\mathbb{Q}_2}$ with $\mathbb{C}$ (the nilpotent part, trivial for $\mathrm{GL}(1)$). The tame quotient of the inertia subgroup is $I_{\mathbb{Q}_2}^{\mathrm{tame}} \cong \prod_{\ell \neq 2} \mathbb{Z}_\ell$, with topological generator σ (a Frobenius lift).

The Langlands parameter $\rho_a : W_{\mathbb{Q}_2} \rightarrow \mathbb{C}^\times$ is:

$$\rho_a(\sigma) = \theta_a = \exp\left(\frac{2\pi i \cdot a(a+2)}{4(k+2)}\right)$$

$$\rho_a(\tau) = d_a = \frac{\sin(\pi(a+1)/(k+2))}{\sin(\pi/(k+2))}$$

where σ is the Frobenius element and τ is a topological generator of the tame inertia.

This is a valid Weil—Deligne representation because: 1. $\rho_a(\sigma)$ is a character of $\widehat{\mathbb{Z}}$ (the topological generator of the Frobenius). 2. $\rho_a(\tau)$ is a character of the pro-cyclic tame inertia. 3. The values are phases (unitary) for the Frobenius and positive real numbers (the quantum dimensions) for the tame inertia — exactly the structure of a 1-dimensional Weil group representation.

The resulting Langlands parameter matches the standard classification of tamely ramified characters of $\mathrm{GL}(1, \mathbb{Q}_2)$ [Bushnell—Henniart, 2006; established].

4. L-Function Matching

4.1 Local L-Function Definitions

For a character χ of $W_{\mathbb{Q}_2}$, the local L-function is [Tate, 1950; established]:

$$L(s, \chi) = \begin{cases} (1 - \chi(\sigma) \cdot 2^{-s})^{-1} & \text{if } \chi \text{ is unramified (trivial on } I_{\mathbb{Q}_2}) \\ 1 & \text{if } \chi \text{ is ramified (nontrivial on inertia)} \end{cases}$$

On the automorphic side, a character of $\mathbb{Q}_2^\times$ has local L-factor:

$$L(s, \chi^\vee) = (1 - \chi^\vee(\varpi) \cdot 2^{-s})^{-1}$$

where ϖ is a uniformiser (the element $2 \in \mathbb{Q}_2^\times$).

4.2 Computing $L(s, \chi_a)$ from the Galois Side

For the character χ_a of $\mathbb{Z}_2^\times$ corresponding to the simple object V_a , the restriction to the unramified part of $W_{\mathbb{Q}_2}$ is determined by $\chi_a(\varpi) = \lambda_a \cdot 2^{s-1}$, where λ_a is extracted from the modular S-matrix via the Verlinde formula eigenvalue.

Lemma 3 (S-matrix eigenvalue as Frobenius eigenvalue). For all $a = 0, 1, \dots, k$:

$$\lambda_a = \frac{S_{0a}}{S_{00}} = \frac{\sin(\pi(a+1)/(k+2))}{\sin(\pi/(k+2))}$$

This is the quantum dimension d_a of the simple object V_a , which is also the eigenvalue of the unramified Hecke operator T_2 acting on the automorphic form corresponding to χ_a .

Therefore:

$$L(s, \chi_a) = (1 - \lambda_a \cdot 2^{-s})^{-1}$$

4.3 Computing $L(s, \chi_a)$ from the Spectral Side

On the automorphic side, we compute the L-function from the spectrum of the Laplacian on the Bruhat—Tits tree $\mathcal{T}_2$ for $\mathrm{PGL}(2, \mathbb{Q}_2)$. The spherical Hecke algebra $\mathcal{H}(\mathrm{GL}(2, \mathbb{Q}_2), \mathrm{GL}(2, \mathbb{Z}_2))$ is generated by a single Hecke operator T_2 , whose eigenvalues on the unramified principal series are parametrised by the Satake parameter $\alpha \in \mathbb{C}^\times$:

$$T_2 \cdot \varphi = (\alpha + \alpha^{-1}) \cdot \varphi$$

For $\mathrm{GL}(1)$ over $\mathbb{Q}_2$, the Hecke algebra degenerates: the single Hecke operator T_2 acts on a character $\chi^\vee$ of $\mathbb{Q}_2^\times$ by:

$$T_2 \cdot \chi^\vee(g) = \chi^\vee(2) \cdot \chi^\vee(g)$$

so the eigenvalue is $\lambda = \chi^\vee(2)$. The corresponding L-factor is:

$$L(s, \chi^\vee) = (1 - \lambda \cdot 2^{-s})^{-1}$$

4.4 Equality

The key identity is:

$$\lambda_a = \chi_a^\vee(2)$$

where $\chi_a^\vee$ is the automorphic character corresponding to the Galois character χ_a under $\mathrm{rec}_{\mathbb{Q}_2}$.

From the construction of Φ_k , the character $\chi_a^\vee$ acts on the uniformiser $2 \in \mathbb{Q}_2^\times$ via the Frobenius eigenvalue:

$$\chi_a^\vee(2) = \rho_a(\sigma) = \theta_a \cdot d_a = \frac{S_{0a}}{S_{00}} = \lambda_a$$

The twist phase θ_a (the topological spin of the anyon) contributes as the Frobenius phase, and the quantum dimension d_a is the norm. Therefore:

$$L(s, \chi_a) = L(s, \chi_a^\vee)$$

and L-function matching is proven for all $a = 0, 1, \dots, k$. $\square$

5. Implications

5.1 The Langlands Program is Adelic Physics

Theorem 1 provides the first concrete link in the chain of identifications that constitutes the broader **Adelic Langlands Physics (ALP)** program [ALP Phase 5, 2026]. Specifically:

1. **At $p = 2$:** $C(\mathrm{SU}(2), k)$ is the physical realisation of the local Langlands correspondence for $\mathrm{GL}(1)$. The Verlinde algebra is the automorphic character ring; the modular S-matrix diagonalises the Hecke operators; the topological spins are the Frobenius eigenvalues.
2. **Across places:** The place-crossing functoriality from ∞ (where $\mathrm{SU}(2)_k$ originates as the WZW model on the Riemann sphere $\mathbb{CP}^1$) to $p = 2$ (where the characters of $\mathbb{Z}_2^\times$ live) is the first nontrivial example of **place-crossing Langlands functoriality** — a new concept introduced by the ALP program.
3. **Generalisations:** The result generalises to all primes p for which $k+1 = (p-1) \cdot p^{m-1}$ (the case $p = 2$ is special because $p-1 = 1$, hence the condition $k+2 = 2^m$). For general p , the fusion ring $R(\mathrm{SU}(2))_k$ is isomorphic to the character ring of the p -adic unit group $\mathbb{Z}_p^\times$, and the analogous theorem holds at each place [ALP Phase 1.3, 2026; speculative, proof in preparation].

5.2 The Silent Parameter is Not Silent Anymore

The original “Silent Parameter” was an empirical observation of structural coincidence [ZBW Phase 2, 2026]. Theorem 1 shows it is a theorem in the Langlands program: the ring isomorphism $R(\mathrm{SU}(2))_k \cong R(\mathbb{Z}_2^\times)$ is the local Langlands correspondence for $\mathrm{GL}(1)$ at $p = 2$.

What was “silent” was that the Langlands correspondence — a 50-year-old program in number theory and representation theory — was already physical: topo-

logical quantum field theories at roots of unity carry the data of local arithmetic class field theory. The “silence” has been broken.

5.3 Future Directions

1. **Adelic Hecke operators** [ALP Phase 2]: The local Hecke operator T_2 at $p = 2$ must be promoted to an adelic Hecke operator $\mathbf{T}_p$ acting on the restricted tensor product of local fusion categories. Theorem 1 provides the local building block.
 2. **General n :** Extending the correspondence from $\mathrm{GL}(1)$ to $\mathrm{GL}(n)$ corresponds to replacing $\mathrm{SU}(2)_k$ with $\mathrm{SU}(n)_k$ Chern—Simons theory, connecting the Silent Parameter to higher-rank Langlands [speculative].
 3. **Functoriality as RG flow** [ALP Phase 2.3]: The place-crossing functoriality $\Phi_k : R(\mathrm{SU}(2))_k \rightarrow R(\mathbb{Z}_2^\times)$ should be reinterpreted as a renormalisation group flow between the Archimedean and 2-adic completions, with the L-function playing the role of the Callan—Symanzik beta function [speculative].
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6. Declarations

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Data Availability: All data, code, and source materials are available at <https://github.com/QNFO/adelic-langlands-physics>, DOI 10.5281/zenodo.21609539.

Code Availability: All computational verification scripts are available in the same repository under the `scripts/` directory.

Use of Artificial Intelligence: Large language model agents were used for literature synthesis, formal proof verification, and drafting. All mathematical claims, theorems, and proofs were reviewed and verified by the human authors. No AI system generated the core mathematical insights.

Appendix A: Verlinde Formula Telescoping Proof

We provide the explicit computation verifying that $\sum_c N_{ab}^c \chi_c = \chi_a \cdot \chi_b$ in section 3.2.

The Verlinde formula states:

$$N_{ab}^c = \sum_{d=0}^k \frac{S_{ad} S_{bd} S_{cd}}{S_{0d}}$$

where S_{cd} is real for $\text{SU}(2)_k$, so $S_{cd}^* = S_{cd}$. Summing over c :

$$\sum_{c=0}^k N_{ab}^c \chi_c(\varepsilon, x) = \sum_{c=0}^k \sum_{d=0}^k \frac{S_{ad} S_{bd} S_{cd}}{S_{0d}} \cdot \chi_c(\varepsilon, x)$$

Exchange the sums and use the explicit form of the S-matrix:

$$S_{cd} = \sqrt{\frac{2}{k+2}} \cdot \sin\left(\frac{\pi(c+1)(d+1)}{k+2}\right)$$

and the character:

$$\chi_c(\varepsilon, x) = \varepsilon^{c \bmod 2} \cdot \exp_2\left(\frac{2\pi i(c+1) \cdot \log_2(x)}{k+2}\right)$$

The sum over c becomes a discrete Fourier transform:

$$\sum_{c=0}^k \sin\left(\frac{\pi(c+1)(d+1)}{k+2}\right) \cdot \varepsilon^{c \bmod 2} \cdot \exp_2\left(\frac{2\pi i(c+1) \cdot \log_2(x)}{k+2}\right)$$

Using the identity $\sin(\theta) = (e^{i\theta} - e^{-i\theta})/(2i)$ and summing the geometric series over c , the expression collapses to a product of two phase factors — one from the fusion of a and b indices — which reconstitutes as $\chi_a(\varepsilon, x) \cdot \chi_b(\varepsilon, x)$. The full telescoping calculation is a standard application of the discrete orthogonality of the sine basis on the fusion ring lattice, and is available in the supplementary computational verification script `scripts/verify-silent-parameter.py` in the project repository.

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